

Solutions to Problem Set 6

1. Observe that $\mathcal{N}(A)$ is a subspace of V^5 . So we must have $\dim(\mathcal{N}(A)) \leq 5$.

(a) The 8×5 zero matrix has $\dim(\mathcal{N}(A)) = 5$, so 5 is the largest value.

(b) The matrix

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

has $\dim(\mathcal{N}(A)) = 0$, so 0 is the smallest value.

2. (a) $\det(A) = -3(2) - 5(2) = -16$

(b) We use cofactor expansion along the first column.

$$\det(A) = 1[-2(1) - 4(1)] + 1[2(4) - 3(-2)] = 8$$

3. (a) Observe that

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 2 \\ 2 & 5 - \lambda \end{pmatrix}.$$

We have

$$\det(A - \lambda I) = (1 - \lambda)(5 - \lambda) - 2(2) = \lambda^2 = 6\lambda + 1.$$

The roots of $\lambda^2 - 6\lambda + 1 = 0$ are $\lambda = 3 \pm \sqrt{8}$. The eigenvalues of A are

$$\lambda_1 = 3 + \sqrt{8}, \quad \lambda_2 = 3 - \sqrt{8}.$$

(b) Since the matrix is lower triangular, the eigenvalues are simply the diagonal entries. Therefore

$$\lambda_1 = 1, \quad \lambda_2 = -5, \quad \lambda_3 = \sqrt{2}.$$

4. To check the invertibility of A we calculate $\det(A)$. Using cofactor expansion along the first row, we find that

$$\det(A) = yz^2 - zy^2 - (xz^2 - zx^2) + xy^2 - yx^2.$$

After a bit of algebra we find that

$$\det(A) = (x - y)(y - z)(z - x).$$

We see that $\det(A) \neq 0$ if and only if x, y, z are distinct. It follows that A is invertible if and only if x, y, z are distinct.

5. A basis for $\mathcal{N}(A)$ is given by

$$\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

There the dimension of $\mathcal{N}(A)$ is 2. The Rank and Nullity Theorem implies that the rank of A is $4 - 2 = 2$.

6. (a) We cannot conclude that x is an eigenvector. For an example, let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and put $\lambda = 1$. Then

$$A - \lambda I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

and we see that $(A - \lambda I)^2 = 0$ so every element of V^2 belongs to $\mathcal{N}((A - \lambda I)^2)$. In particular

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

is an element of $\mathcal{N}((A - \lambda I)^2)$ but x is not an eigenvector of A .

(b) We can conclude that λ is an eigenvalue of A . Since $\mathcal{N}((A - \lambda I)^2)$ is nontrivial, we know that

$$\det((A - \lambda I)^2) = 0.$$

Since the determinant of a product is the product of the determinants, we conclude that

$$\det(A - \lambda I) = 0,$$

and this implies that λ is an eigenvalue of A .